

Vanishing Viscosity for Smooth Transport

Partial Differential Equations

Homework 4, Exercise 14

Selected solution, edited for standalone reading

Let $\mathbb{T}^d = \mathbb{R}^d / (2\mathbb{Z})^d$ be the periodic box and let $b \in C^1(\mathbb{T}^d; \mathbb{R}^d)$ be time-independent with $\operatorname{div} b = 0$. For common initial data $u_0 \in H^1(\mathbb{T}^d)$, compare

$$\begin{aligned}\partial_t u^\epsilon + b \cdot \nabla u^\epsilon - \epsilon \Delta u^\epsilon &= 0, & \partial_t \bar{u} + b \cdot \nabla \bar{u} &= 0, \\ u^\epsilon(\cdot, 0) &= \bar{u}(\cdot, 0) = u_0, & 0 \leq t \leq T.\end{aligned}$$

The solutions are understood in the energy class, with $u^\epsilon \in C([0, T]; H^1) \cap L^2(0, T; H^2)$ and $\bar{u} \in C([0, T]; H^1)$. The aim is to bound their L^2 difference by $C\sqrt{\epsilon}$, uniformly on $[0, T]$.

The differential identities below can first be established for smooth data; approximation then yields the same estimates in the stated energy class. The Laplacian of $\bar{u}$ in the difference equation is interpreted weakly.

Energy-estimate proof

1. Equation for the difference

Let $v^\epsilon = u^\epsilon - \bar{u}$. Subtracting the equation for $\bar{u}$ from the equation for u^ϵ , we find that v^ϵ satisfies:

$$\partial_t v^\epsilon + b \cdot \nabla v^\epsilon - \epsilon \Delta u^\epsilon = 0 \quad \text{in } \mathbb{T}^d \times (0, T]. \quad (1)$$

Adding and subtracting $\epsilon \Delta \bar{u}$, we can rewrite this in terms of v^ϵ :

$$\partial_t v^\epsilon + b \cdot \nabla v^\epsilon - \epsilon \Delta v^\epsilon = \epsilon \Delta \bar{u}. \quad (2)$$

The initial condition is $v^\epsilon(\cdot, 0) = u_0 - u_0 = 0$.

2. Energy estimate for the difference

Pair equation (2) with v^ϵ and integrate over the torus. Periodicity removes boundary terms, and incompressibility cancels the transport contribution:

$$\int_{\mathbb{T}^d} (b \cdot \nabla v^\epsilon) v^\epsilon dx = -\frac{1}{2} \int_{\mathbb{T}^d} (\operatorname{div} b) (v^\epsilon)^2 dx = 0.$$

Integrating the diffusion and forcing terms by parts gives

$$\frac{1}{2} \frac{d}{dt} \|v^\epsilon\|_{L^2}^2 + \epsilon \|\nabla v^\epsilon\|_{L^2}^2 = -\epsilon \int_{\mathbb{T}^d} \nabla \bar{u} \cdot \nabla v^\epsilon dx.$$

Using the Cauchy-Schwarz inequality and Young's inequality ($ab \leq \frac{a^2}{2} + \frac{b^2}{2}$) on the right side:

$$-\epsilon \int_{\mathbb{T}^d} \nabla \bar{u} \cdot \nabla v^\epsilon dx \leq \epsilon \|\nabla \bar{u}\|_{L^2} \|\nabla v^\epsilon\|_{L^2} \leq \frac{\epsilon}{2} \|\nabla \bar{u}\|_{L^2}^2 + \frac{\epsilon}{2} \|\nabla v^\epsilon\|_{L^2}^2.$$

Absorbing $\frac{\epsilon}{2} \|\nabla v^\epsilon\|_{L^2}^2$ into the left-hand side, we get:

$$\frac{1}{2} \frac{d}{dt} \|v^\epsilon\|_{L^2}^2 + \frac{\epsilon}{2} \|\nabla v^\epsilon\|_{L^2}^2 \leq \frac{\epsilon}{2} \|\nabla \bar{u}\|_{L^2}^2.$$

Dropping the non-negative term $\frac{\epsilon}{2} \|\nabla v^\epsilon\|_{L^2}^2$, we have:

$$\frac{d}{dt} \|v^\epsilon(\cdot, t)\|_{L^2}^2 \leq \epsilon \|\nabla \bar{u}(\cdot, t)\|_{L^2}^2. \quad (3)$$

3. Gradient bound for the transport solution

To bound the right side of (3), we need an estimate for $\|\nabla \bar{u}\|_{L^2}$. Taking the spatial gradient ∇ of the transport equation $\partial_t \bar{u} + b \cdot \nabla \bar{u} = 0$, we get:

$$\partial_t(\nabla \bar{u}) + b \cdot \nabla(\nabla \bar{u}) + (Db)^T \nabla \bar{u} = 0,$$

where $Db = (\partial_j b_i)_{i,j}$ is the Jacobian matrix. Its operator norm is denoted by $\|Db\|_{L^\infty}$. Multiply this by $\nabla \bar{u}$ and integrate over $\mathbb{T}^d$:

$$\frac{1}{2} \frac{d}{dt} \|\nabla \bar{u}\|_{L^2}^2 + \int_{\mathbb{T}^d} (b \cdot \nabla(\nabla \bar{u})) \cdot \nabla \bar{u} \, dx + \int_{\mathbb{T}^d} ((Db)^T \nabla \bar{u}) \cdot \nabla \bar{u} \, dx = 0.$$

Again, the transport term vanishes because $\operatorname{div} b = 0$. Thus,

$$\frac{1}{2} \frac{d}{dt} \|\nabla \bar{u}\|_{L^2}^2 = - \int_{\mathbb{T}^d} ((Db)^T \nabla \bar{u}) \cdot \nabla \bar{u} \, dx \leq \|Db\|_{L^\infty} \|\nabla \bar{u}\|_{L^2}^2.$$

By Grönwall's inequality, for any $t \in [0, T]$:

$$\|\nabla \bar{u}(\cdot, t)\|_{L^2}^2 \leq \|\nabla u_0\|_{L^2}^2 e^{2\|Db\|_{L^\infty} t}. \quad (4)$$

4. Convergence rate

Integrate equation (3) from 0 to t (recall $\|v^\epsilon(\cdot, 0)\|_{L^2} = 0$):

$$\begin{aligned} \|v^\epsilon(\cdot, t)\|_{L^2}^2 &\leq \epsilon \int_0^t \|\nabla \bar{u}(\cdot, s)\|_{L^2}^2 \, ds \\ &\leq \epsilon \int_0^t \|\nabla u_0\|_{L^2}^2 e^{2\|Db\|_{L^\infty} s} \, ds \\ &\leq \epsilon t \|\nabla u_0\|_{L^2}^2 e^{2\|Db\|_{L^\infty} t}. \end{aligned}$$

Taking the supremum over $t \in [0, T]$ and then the square root gives:

$$\sup_{t \in [0, T]} \|u^\epsilon(\cdot, t) - \bar{u}(\cdot, t)\|_{L^2} \leq C \epsilon^{1/2},$$

where $C = \sqrt{T} \|\nabla u_0\|_{L^2} e^{\|Db\|_{L^\infty} T}$. Note that this constant C depends only on T , $\|Db\|_{L^\infty}$, and $\|u_0\|_{H^1}$, as required.

Finally, as $\epsilon \rightarrow 0$, $C \epsilon^{1/2} \rightarrow 0$. Therefore, it immediately follows that:

$$u^\epsilon \rightarrow \bar{u} \quad \text{in } L^\infty([0, T]; L^2(\mathbb{T}^d)).$$

□